

Literature-Implied Partial Subcase for arXiv:1403.6737

FA/Banach run fa_banach_001

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Status

This is a `literature_implied_answer` (partial subcase) packet. It does not claim a new proof of Pisier’s general analyticity conjecture. It records that a later theorem of Arhancet [2] gives a stronger Fourier-multiplier subcase than the one proved in the 2014 source paper [1].

Source Problem

The source paper recalls Pisier's conjecture as follows. If X is a K-convex Banach space and $(T_t)_{t \geq 0}$ is a weak-star continuous semigroup of selfadjoint contractive operators on $L^\infty(\Omega)$, where Ω is a measure space, then the conjecture predicts that analyticity of the scalar $L^p(\Omega)$ semigroup is preserved after tensoring with Id_X on the Bochner space $L^p(\Omega, X)$.

$$\bigcup_{k=1}^{\infty} J_{\Omega} \quad \text{and} \quad J_{\Omega} \cap \bigcap_{k=1}^{\infty} J_{\Omega_k} = \emptyset.$$

induces a bounded operator on the Bochner space $L^2(\Omega, X)$ where Ω is a probability space and where $\varepsilon_1, \varepsilon_2, \dots$ is a sequence of independent random variables with $P(\varepsilon_k = 1) = P(\varepsilon_k = -1) = \frac{1}{2}$. Such a Banach space X is called K-convex. The heart of his proof relies on the fact, proved by himself in his article, that if X is a K-convex Banach space then any w^* -continuous semigroup $(T_t)_{t \geq 0}$ of positive unital selfadjoint Fourier multipliers on a locally compact abelian group G induces a strongly continuous bounded analytic semigroup $(T_t \otimes Id_X)_{t \geq 0}$ of contractions on the Bochner space $L^p(G, X)$ for any $1 < p < \infty$. In 1981, in the seminars [Pis1] and [Pis2] which announced the results of his paper, he stated several natural questions raised by his work. In particular, he conjectured [Pis1, page 17] that the same property holds for any w^* -continuous semigroup $(T_t)_{t \geq 0}$ of selfadjoint contractive operators on $L^\infty(\Omega)$ where Ω is a measure space (see also the recent preprint [Xu, Problem 11] for a more general question). Note that it is well-known [Ste, III2 Theorem 1] that such a semigroup induces a strongly continuous bounded analytic semigroup of contractions on the associated L^p -space $L^p(\Omega)$ and the conjecture says that the property of analyticity is preserved by the tensorisation of the identity Id_X of a K-convex Banach space X .

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Figure 1: Source evidence from arXiv:1403.6737, page 1: the paragraph recording Pisier’s conjecture on tensorized analyticity for K-convex target spaces.

The source paper gives partial positive results for Fourier multipliers on compact abelian groups under an OK-convex operator-space realization hypothesis. The packet below records a later, separate strengthening of that partial route.

Supporting Literature

The supporting paper arXiv:1807.00875 introduces the following condition: a Banach space X is K' -convex if it is isomorphic to a Banach space E with an operator space structure such that the vector-valued Schatten space $S^p(E)$ is K -convex. In operator-space language, E is OK-convex. The paper notes that every K' -convex space is K -convex, and conjectures that the two classes coincide.

Its Corollary 14 gives the locally compact abelian Fourier-multiplier subcase.

Definition 12 *We say that a Banach space X is K' -convex if X is isomorphic to a Banach space E such that there exists an operator space structure on E such that the vector-valued Schatten space $S^p(E)$ is K -convex.*

With the language of operator spaces, the condition on E means that the operator space E is OK-convex. In the situation of this definition, E is a closed subspace of $S^p(E)$, hence a K -convex Banach space. Since K -convexity is preserved under isomorphisms, we deduce that a K' -convex Banach space is necessarily K -convex. We conjecture the following.

Conjecture 13 *The class of K' -convex Banach spaces coincides with the class of K -convex Banach spaces.*

It might even be possible that each K -convex Banach space admits an OK-convex operator space structure. In Section 5, we will show that the class of K' -convex Banach spaces shares the same nice properties of stability under usual operations than the class of K -convex Banach spaces. This is an observation that goes in the direction of the conjecture.

Now, we state the consequence for abelian locally compact groups.

Corollary 14 *Suppose that G is a second countable abelian locally compact group. Let $(T_t)_{t \geq 0}$ be a weak* continuous semigroup of (not necessarily positive) selfadjoint contractive Fourier multipliers on $L^\infty(G)$. Let X be a K' -convex Banach space. Consider $1 < p < \infty$. Then $(T_t)_{t \geq 0}$ induces a strongly continuous bounded analytic semigroup $(T_t \otimes \text{Id}_X)_{t \geq 0}$ of contractions on the Bochner space $L^p(G, X)$.*

Proof. Note that we have a p -isomorphism of von Neumann algebras between $L^\infty(G)$ and

Figure 2: Supporting evidence from arXiv:1807.00875, page 10: Definition 12, Conjecture 13, and Corollary 14 for second-countable abelian locally compact groups.

Idea of the Proof

The source conjecture asks whether tensoring by Id_X preserves bounded analyticity for selfadjoint contractive semigroups when X is K -convex. The 2018 theorem proves this conclusion for a structured but substantial subcase: the underlying measure space is a second-countable locally compact abelian group, the semigroup consists of Fourier multipliers, and X satisfies the stronger K' -convex/OK-convex realization hypothesis. Because K' -convexity implies K -convexity, this is a genuine subcase of the source conjecture, not a change of target.

Theorem 1. *Let G be a second-countable abelian locally compact group, let $1 < p < \infty$, and let X be a K' -convex Banach space. Suppose $(T_t)_{t \geq 0}$ is a weak-star continuous semigroup of selfadjoint contractive Fourier multipliers on $L^\infty(G)$. Then $(T_t \otimes \text{Id}_X)_{t \geq 0}$ is a strongly continuous bounded analytic semigroup of contractions on $L^p(G, X)$.*

Proof. This is exactly Corollary 14 of Arhancet [2]. The hypotheses there are: G is second countable, abelian, and locally compact; $(T_t)_{t \geq 0}$ is weak-star continuous and consists of selfadjoint con-

tractive Fourier multipliers on $L^\infty(G)$; X is K' -convex; and $1 < p < \infty$. The conclusion is precisely that the tensorized semigroup $(T_t \otimes \text{Id}_X)_{t \geq 0}$ induces a strongly continuous bounded analytic semigroup of contractions on the Bochner space $L^p(G, X)$.

Definition 12 and the following paragraph in [2] explain that K' -convexity means that X is isomorphic to a Banach space with an OK-convex operator space structure, and that such spaces are K-convex. Hence the theorem is a positive instance of Pisier's K-convex analyticity prediction, restricted to this Fourier-multiplier/locally compact abelian setting. $\square$

Verification and Limitations

No computational verification is involved. The verification is the direct identification between the source conjecture and the supporting corollary:

- $L^\infty(G)$ is a scalar commutative von Neumann algebra attached to the Haar-measure space G ;
- Fourier multipliers in Corollary 14 are a subclass of the selfadjoint contractive operators appearing in the source conjecture;
- K' -convexity is a sufficient extra hypothesis implying K-convexity;
- the conclusion is exactly bounded analyticity after tensoring with Id_X .

The result does not solve Pisier's general conjecture for arbitrary measure spaces, arbitrary selfadjoint contractive semigroups, or all K-convex target spaces. The supporting paper explicitly keeps the equality of K-convex and K' -convex spaces as a conjecture.

Search Bounds

The cheap run indexes were searched for 1403.6737, 1807.00875, Pisier, analyticity of semigroups, K-convex, OK-convex, Fourier multipliers, Ritt operators, and related terms. No prior packet or attempt for this exact subcase was found. A nearby arXiv:1407.1142 packet concerns a different positive/contractive UMD-Banach-lattice regularity cluster.

External arXiv/web search on June 14, 2026 found arXiv:1807.00875 as the relevant later strengthening. No full solution of the general conjecture was identified within this bounded check.

Human Review Recommendation

Review as a literature-status partial subcase. In particular, verify that K' -convexity is not being conflated with K-convexity and that the Fourier-multiplier/locally compact abelian restriction is stated everywhere.

References

- [1] C. Arhancet, *On a conjecture of Pisier on the analyticity of semigroups*, arXiv:1403.6737.
- [2] C. Arhancet, *On analyticity of semigroups on Bochner spaces and on vector-valued noncommutative L^p -spaces*, arXiv:1807.00875.